DISCHARGE SUMMARY

DEPARTMENT OF CLINICAL HEMATOLOGY AND MEDICAL ONCOLOGY,
POSTGRADUATE INSTITUTE OF MEDICAL EDUCATION AND RESEARCH
CHANDIGARH – 160 012 (INDIA)

NAME	Vipan Kumar Garg	Age/ Sex	68/M	C.R. No.	201405136462
BSA:1.91m2	Ht:180.34cm Wt:73kg	DOA	27/06/26	Admission No	2026053678
Consultant	Dr Pankaj Malhotra	DOD	15/07/26	Phone No	9815311118
DIAGNOSIS	Relapse/ Refractory IgG kappa multiple myeloma with clinical relapse ISS-stageII, R-ISS-stageIII, IMWG- Standard risk diseases FISH: 3,4 copies of both 1q21 and 1p32 loci in 10% cells Dara-VRd→Dara-VCd→Bortezomib maintenance with lenalidomide- induced DRESS syndrome with carfilzomib-induced thrombotic microangiopathy resulting in CKD TECLISTAMAB : **30/06/2026:** Step-up dose 1 – **4.5 mg (0.06 mg/kg)** **03/07/2026:** Step-up dose 2 – **22.5 mg (0.3 mg/kg)** **07/07/2026:** First full dose – **112.5 mg (1.5 mg/kg)** IGRA positive, Isoniazid-rifampicin (HR) prophylaxis was initiated for latent tuberculosis.				

Chief Complaints:

- Admitted for Teclistamab

BGH and HOPI:

The patient was apparently well until April-May 2022, when he developed persistent back pain and right-sided chest pain, for which he was evaluated at PGIMER, Chandigarh. Initial workup revealed markedly elevated serum free light chains with free kappa 857.95 mg/L, free lambda 17.82 mg/L and Kappa/ Lambda ratio of 48.15 (30/05/2022) along with 24-hour urine protein of 922 mg/day (01/06/2022). Peripheral blood smear showed mild anisocytosis with normocytic normochromic anemia without circulating plasma cells or rouleaux formation. Bone marrow examination and serum protein electrophoresis confirmed the diagnosis of IgG Kappa Multiple Myeloma with significant skeletal involvement fulfilling CRAB criteria due to extensive bone disease. MRI dorsal spine dated 25/07/2022 demonstrated multiple enhancing vertebral lesions with pathological collapse of D1 and D6 vertebrae, associated paravertebral soft tissue component with extradural extension and foraminal compression, along with bilateral costochondral involvement. FISH analysis revealed gain of chromosome 1q21 (CKS1B amplification) with 3–4 copies in approximately 10% of cells, while del17p, t(4;14), t(14;16), t(14;20) and t(11;14) were negative.

The patient was started on DaraVCd (Daratumumab, Bortezomib, Cyclophosphamide and Dexamethasone) in June 2022 and completed five cycles by October 2022 along with bisphosphonate therapy. Lenalidomide was initiated initially but subsequently discontinued due to thrombocytopenia and replaced with cyclophosphamide. Response assessment showed progressive reduction in disease burden with M-spike decreasing to 0.06 g/dL on 18/10/2022 and 0.04 g/dL on 16/02/2023, while SPEP on 23/05/2023 showed no detectable M-spike. The patient achieved MRD-negative complete remission in December 2022, following which the planned autologous stem cell transplantation was deferred. He subsequently received maintenance bortezomib from January 2023 to August 2023 along with periodic zoledronic acid therapy. Repeat PET-CT and MRD assessment in August 2023 remained negative for active disease.

During follow-up, biochemical relapse was suspected with rising serum free light chains, with serum kappa increasing to approximately 98 mg/L by January 2024 and reappearance of M-spike measuring 0.24 g/dL on 30/12/2023. Bone marrow examination performed on 10/02/2024 showed normocellular marrow with trilineage hematopoiesis and approximately 3% plasma cells without clustering, while PET-CT dated 13/02/2024 demonstrated metabolically active lytic lesions involving ribs and thoracic vertebrae consistent with residual/recurrent disease, persistent collapse of D1 and D2 vertebrae, along with infective/inflammatory changes in the left lung, mediastinal and supraclavicular lymphadenopathy, hepatomegaly and prostatomegaly. In view of relapsed disease, the patient was initiated on Carfilzomib, Daratumumab and Pomalidomide-based therapy with zoledronic acid in March 2024.

In September 2024, the patient developed acute kidney injury with anuria and serum creatinine of 5.54 mg/dL, necessitating hemodialysis. Evaluation revealed a positive direct Coombs test with associated anemia and thrombocytopenia, and HUS/TMA was considered in the differential diagnosis. Ultrasonography of the abdomen performed on 08/09/2024 demonstrated bilateral increased renal parenchymal echotexture, while echocardiography on 09/09/2024 showed preserved left ventricular systolic function with LVEF 65% and Grade I diastolic dysfunction. Sputum analysis was positive for Streptococcus pneumoniae, and HRCT chest showed minimal bilateral pleural effusion with bibasal consolidative changes.

Subsequent disease monitoring demonstrated improvement and stabilization of serum free light chains with free kappa 20.31 mg/L, free lambda 7.25 mg/L and Kappa/Lambda ratio 2.80 on 29/11/2024, followed by free kappa 46.57 mg/L, free lambda 15.04 mg/L and ratio 3.09 on 10/02/2025, and free kappa 48.70 mg/L, free lambda 19.25 mg/L with ratio 2.53 on 24/05/2025. Serum immunofixation electrophoresis during this period demonstrated only a faint

residual IgG kappa monoclonal band. Since January 2025, the patient has been receiving maintenance therapy with monthly Daratumumab, Pomalidomide 2 mg (21 days on and 7 days off), and Denosumab every three months.He received Cycle 20 Day 1 of maintenance Daratumumab along with Denosumab on 21/08/2025

The patient is currently admitted in the private ward for administration of Teclistamab therapy with planned step-up dosing as per protocol and close monitoring for treatment-related adverse events including cytokine release syndrome (CRS), ICANS, cytopenias, and infections

Past History:

- H/o lenalidomide induced DRESS
- H/o carfilzomib induced TMA (CKD)
- H/o Asthma since 2003 (on oracort 200 inhaler + Montair LC regularly)
- H/o Hypertension since 2018 on med Telmisartan (40mg)
- H/o pre-diabetic in Jan 2022 (Hb1AC - 6)
- H/o thyroid since 2018 on medicine
- H/o prostate on Alfuzosin (10mg) daily

Personal History

- Sleep: Normal pattern, duration
- Bladder: Normal
- Bowel: Normal
- Diet: Vegetarian
- Appetite: Normal
- Addictions: None

Family history:

- No h/o similar illness in family.
- No h/o malignancy/ tuberculosis/ recent fever in family.

GENERAL EXAMINATION	SYSTEMIC EXAMINATION
On admission: Patient was conscious, cooperative and oriented to time, place and person. BP: 136/82mm Hg PULSE:84/min RR: 20/min Sp02: 98% at RA Temperature: Afebrile P+I+ C-Cl-L+E+ No petechiae / purpura	CVS: S1, S2 heard. No murmurs RS: B/L NVBS, air entry equal bilaterally. No added sounds. Abdomen: soft, non tender , no hepatosplenomegaly CNS: HMF intact. No focal deficits, B/L UL power 5/5, B/L LL power 5/, tone normal.

INVESTIGATIONS:

	27/06	**30/06**	**01/07**	**02/07**	**03/07**	**04/07**	**06/07**	**07/07**	**09/07**	**10/07**	**11/07**
Hb	11.3	11.1	11.1	10.7	10	11.4	12	12.9	11.5	11.6	11.5
MCV	96.6	103	97.7	95.5	98.4	95.1	95.6	95.7	95.3	95.3	96.9
TLC	7.5	6.42	8.3	4.18	4.12	8.6	4.9	6	6.01	6.47	6.1
ANC	-	4.47	7.8	4.08	4.24	7.7	3.8	4.1	5.08	5.12	.
PLT	152	139		111	98	108	102	102	107	107	100
Na	137.2	135.9	136.9	137.2	145.6	134.2	138	136.8	136.1	135.4	135.2
K	4.5	5.11	4.65	4.46	3.67	4.51	3.97	4.3	4.1	4.05	4.23

Ca	7.96	7.92	8.56	7.68	8.9	8.48	9.14	9.62	8.69	8.33	7.96
Mg	2.18	2.34	2.07	1.9	1.8	1.92	1.92	1.96	1.89	1.83	1.88
P	2.37	1.95	2.2	1.75	1.98	2.64	3.5	4.08	3.76	3.07	2.66
Urea	35.8	29.1	40.9	57.4	56.7	52	62.7	62.1	76.5	60.4	52.9
Creat	0.2	1.44	1.71	1.89	1.7	1.68	2.17	2.16	2.47	2.12	1.86
OT	18.7	18.4	25.5	12.4	13.45	17.5	17.1	14.9	14.4	15.4	15
PT	13.9	5.9	12.2	9.1	9.42	12.5	13.8	15.9	8.9	9.53	9.7
ALP	95	98	88	76	70	90.3	81.2	91	104	102	94
TB	0.2	0.17	0.196	0.2	0.256	0.206	0.2	0.63	0.79	0.601	0.45
CB	0.09	0.06	0.043	0.09	0.042	0.131	0.091	0.28	10.5	0.319	0.25
TP	6.41	6.29	6.2	5.79	5.78	6.13	5.92	6.97	6.41	5.8	5.69
Alb	4.36	4.2	4.14	4.06	3.87	4.27	4.13	4.65	4.36	4.12	3.93
UA	7.8	6.8	6.57	7.7	7.2	7	8.39	8.7	10.5	9.81	8.9
CRP	1.78	1.32	0.956	9.04	9.08	1.67	2.15	0.65	0.62	0.487	0.53
LDH	152	188	257	135	120	167	173	178	159	148	166
Ferritin	-		119	65.7	62.4	60.5	-	52.3	50.3	53.6	54.1

ICE SCORE	**27/06**	**30/06**	**01/07**	**02/07**	**03/07**	**04/07**	**06/07**	**07/07**	**09/07**	**10/07**	**11/07**
MORNING	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10

EVENING	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10

Cardiac markers (02/07/26):

- Pro BNP: 407; Trop T: 14.7

Baseline FISH: MM-145/22: 3-4 copies each of 1q21(CKS1B locus) and 1p32(CDKN2C locus) seen in 10% of cells, suggestive of trisomy/tetrasomy of chromosome

Date	LDH	M-spike	SFLC	24 hr UP	Ig profile	B2 Mi	BM	MRD	PET
01/06/22	247	1.74g/dl	kappa free chain :857.95 lambda free chain:17.82 25/07/22: freekappa: 16.97 free lambda: 11.7 ratio: 1.45	922.0	IgA:54 IgG:259 IgM:29	4.806	hypercellular BM 15% plasma cells	-	13/06/22multiple non- hypermetabolic to hypermetabolic lytic lesions.
29/09/22 (post 5 week chemo)		0.22g/dl	free kappa: 16.97 free lambda: 11.73 Ratio: 1.4500						
14/12/22							Cellularity 30-40% and 2% plasma cells	MRD-828/22: Nil	
Feb 2023		0.04g/dl	free kappa: 8.08 free lambda:4.19 Ratio: 1.93						

May 2023		no M spike	free kappa:16.65 free lambda:13.74 Ratio:1.21						
aug 2023		very faint band	free kappa:23.08 free lambda:12.71 Ratio: 1.82					MRD- 643/23:Ni	
Dec,2023		0.24g/dl	free kappa:56.82 free lambda:17.28 Ratio: 3.29						
Jan2024		0.2g/dl	free kappa:92.94 free lambda:18.35 Ratio:5.065						
feb,2024									13/02/24 1.Metabolically active lytic lesions involving few skeletal sites as described (pathological fractures involving right 6th & left 8th ribs, collapsed D6 vertebra) - residual / recurrent neoplastic disease involvement. 2.Non-metabolic lytic lesions in multiple other skeletal sites as described above; D1 & D2 vertebral collapse with no significant metabolic activity.
4/12/24		no M spike(?0.2)	free kappa:20.31 free lambda:7.25 Ratio:2.801	0.14	IgA:<33.0 IgG:684.00 IgM:<21.0				
10/02/25	201	?0.2	free kappa:46.57 free lambda:15.04 Ratio:3.096		IgA:52.7 IgG:736 IgM:<21	7864			

21/08/25	146		free kappa:38.71 free lambda:16.58 Ratio:2.335		IgA:59 IgG:515 IgM:<21	10244			
04/01/26		0.11g/dl	free kappa:37.53 free lambda:16.98 Ratio:2.21						23/03/26 FDG avid right paravertebral soft tissue component at D5-D7 vertebral level and non FDG avid lytic lesions.- residual diseases
15/04/26		?0.2	free kappa:97.65 free lambda:22.36 Ratio:4.367		IgA:88.4 IgG:752 IgM:<21				
24/04/26							24/04/26 cellularity 30-40% BM show 2% plasma cells	0.001% Positive for residual myeloma cells. The myeloma cells are positive for CD38, CD56, CD117, CD81 (dim), CD19 (dim), cytoKappa, negative for CD38, CD27, CD45, cytolambda	
15/04/26		?0.2	free kappa:97.65 free lambda:22.36 Ratio:4.367		IgA:88.4 IgG:752 IgM:<21				

HRCT THORAX (08/09/24):

FINDINGS:

- Minimal bilateral pleural effusion is seen with fissural extension, passive atelectasis and mild smooth interlobular septal thickening of the underlying lung parenchyma.
- Few tiny focal areas of airspace consolidations are seen in basal segments of bilateral lower lobes (right > left.
- The visualized thoracic esophagus is distended with fluid contents.
- Rest of the lung parenchyma is normal in attenuate or
- The tracheobronchial tree is normal
- The heart, great vessels, trachea and mainstem bronchi are normal Minimal strips of pericardial effusion are seen.
- No enlarged mediastinal lymphadenopathy is seen.
- Degenerative changes are seen in the visualized spine.
- Atherosclerotic wall calcifications are seen in the aorta and its branches.

IMPRESSION:

• Minimal bilateral pleural effusion with minimal aspiration changes in bilateral basal segments.

2D echo (09/09/26):

- Normal LV Systolic function, LVEF - 65%
- No RWMA at rest
- Type 1 Diastolic Dysfunction
- Trivial Mitral Regurgitation, No MS
- Sclerodegenarative Aortic valve
- No AS/AR
- Trivial Tricuspid Regurgitation, ERVSP - 30 mmHg
- IVC - 9-10mm, collapsing
- IAS/IVS - Intact
- No e/o Clot/thrombus/Vegetation on TTE
- No Pericardial Effusion

Final Impression

- Normal LV Systollc Functlon, (LVEF - 65%)
- Type I Diastolic Dysfunction

X-ray chest AP (09/09/24): ill defined opacity seen in right mid-lower and left lower zone with obliteration of both CP angles.

USG abdomen and pelvis (08/09/24):

LIVER: Visualised Liver is normal in size, shape and echo texture. No e/o dilated THBR or portal venous radical. visualised Upper CBD and portal vein are normal in caliber.

GALL BLADDER: Gall Bladder is partially distended and shows minimal sludge within the lumen. No abnormal wall thickening noted.

PANCREAS: pancreas and midline structure are obscure due to bowel gas.

SPLEEN: Spleen is normal in size & echo texture. No focal or diffuse lesion is seen

KIDNEYS:

- Right kidney: 11 x 4.9 cm
- Left kidney: 11 x 5 cm
- Both kidneys are normal in shape, size and position.
- Both kidneys show mild diffuse increased renal parenchymal echotexture suggestive of likely acute renal parenchymal disease.

U. BLADDER: Urinary bladder is empty with a foley bulb noted within the lumen.

Prostate appears mildly enlarged in size 42 cc.

Mild free fluid noted within the interbowel region.

IMPRESSION:

• Gall Bladder is partially distended and shows minimal sludge within the lumen. No abnormal wall thickening noted.
• Both kidneys show mild diffuse increased renal parenchymal echotexture suggestive of likely acute renal parenchymal disease.
• Prostate appears mild enlarged.
• Mild free fluid noted within the interbowel region.

SURGICAL PATHOLOGY REPORT (04/11/24): S-30929/2024

Micro :

- Slides and section examined from block show native renal biopsy which is evaluated using H & E, and MT stains under light microscopy. Per level of the sections shows single linear core of renal cortex comprising of a maximum of 10 glomeruli. Out of which 1 glomerulus is globally sclerosed. Rest of the glomeruli show ischemic changes with underlying tuft showing reduplication of basement membrane, mesangial expansion with foaminess and endothelial swelling indicating subacute thrombotic microangiopathy changes.

There is moderate to severe acute tubular injury and presence of vacuolations within tubular epithelial cells accompanied by moderate increase in lymphomononuclear inflammation admixed with an occasional scattered

eosinophils and collection of histiocytes forming granuloma at places. In addition, there is diffuse interstitial edema and early fibrosis with tubular atrophy of 25-30% of the core.
Most of the arteries and arterioles show moderate to severe fibrointimal thickening, medial hypertrophy with mucinoid cystic changes and moderate to severe sclerosis causing obliteration of lumen indicating subacute to chronic thrombotic microangiopathy changes.
DIF for Kappa and Lambda performed on pronase treated paraffin section and show 1-2+ deposits in tubular cast. There is no light chain restriction.
EM sample : Not sent
In this 66 years old male, known case of MM , post Carfilzomib induced TMA,
This native kidney biopsy (slides and block for review) shows features of 1) Evidence of Subacute glomerular and subacute to chronic vascular thrombotic microangiopathy. 2) Moderate to severe Acute tubular injury. 3) Toxic Tubulopathy
Diagnosis:

- kidney : subacute to chronic glomerular and vascular- thrombotic microangiopathy
- kidney: moderate to severe- acute tubular injury
- kidney: toxic-tubulopathy

Pre- TECLISTAMAB workup (JULY,2026)	
IGRA	**positive**
CBC	11.3/7.50/152
IgG	756.0
HbA1c	5.30%
Hep B core antigen	3.13
HCV IgG/IgM	NR/NR
Hep B core IgM	NR
HBsAg	NR

COURSE AND MANAGEMENT:

Mr. Vipan Kumar Gurg is a known case of relapsed/refractory IgG kappa multiple myeloma with clinical relapse. He has ISS stage II, R-ISS stage II, and IMWG standard-risk disease. FISH analysis demonstrated 3–4 copies of both the 1q21 and 1p32 loci in 10% of cells. He had previously received Dara-VRd followed by Dara-VCd and bortezomib maintenance, achieving a progression-free survival of 15 months. Subsequently, he received KPd followed by daratumumab-pomalidomide maintenance. He has a history of lenalidomide-induced DRESS syndrome and carfilzomib-induced thrombotic microangiopathy with chronic kidney disease.

Evaluation at relapse revealed FDG-avid lytic lesions with paravertebral soft tissue involvement on PET-CT (March 2026). SPEP showed an M-protein of approximately 0.2 g/dL, serum free light chain ratio (kappa:lambda) was 97:22, and bone marrow MRD was positive at 0.012%.

He was admitted for initiation of teclistamab under close inpatient supervision. He received teclistamab step-up dose 1 (4.5 mg, 0.06 mg/kg) on 30/06/2026, step-up dose 2 (22.5 mg, 0.3 mg/kg) on 03/07/2026, and the first full dose (112.5 mg, 1.5 mg/kg) on 07/07/2026. He tolerated treatment well without evidence of cytokine release syndrome or immune effector cell-associated neurotoxicity syndrome and remained clinically stable throughout his hospital stay.

His interferon-gamma release assay (IGRA) was positive, and he was initiated on isoniazid-rifampicin (HR) prophylaxis for latent tuberculosis. He was discharged in a stable condition with advice to continue weekly teclistamab therapy and follow up in the Hematology outpatient clinic for further management.

JR: Dr.Sayan, Dr. Jyotiraditya
SR:Dr.Richa
Consultant: Prof Pankaj Malhotra, Dr Aditya Jandial